

Monte Carlo v. DISORT benchmark for Henyey-Greenstein cloud layer

Validation of https://github.com/sepraca/mc_cloud_rt_visualization Monte Carlo (MC) cloud RT code against DISORT for a homogeneous plane-parallel cloud, no atmospheric gas or aerosol extinction, Henyey-Greenstein (HG) phase function, and a Lambertian reflecting surface.

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I. Numerical Implementations

A. MC implementation

https://github.com/sepraca/mc_cloud_rt_visualization

Settings:

- $N_{\text{photons}} = 10^5, 10^6$
 - $1\text{-}\sigma$ MC noise ≈ 0.002 (10^5 photons) to 0.0005 (10^6 photons) absolute for $R \sim 0.5$, assuming binomial counting statistics [$\sigma^2 = R(1-R)/N_{\text{photons}}$]
- To prevent significant cloud side photon escape:
 - Horizontal cloud extent (optical path units) = 150
 - Cloud-base to surface = 0.05 km
- Other optical parameters: as specified in test cases

B. DISORT implementation

PythonicDISORT (Ho, JOSS 2024) v1.7, a Python reimplement of the Stamnes et al. (1988) discrete-ordinates algorithm, validated by its author against Fortran DISORT to $\sim 10^{-6}$ in fluxes.

<https://www.theoj.org/joss-papers/joss.06442/10.21105.joss.06442.pdf>

Settings:

- NQuad = 48 streams, 96 HG Legendre coefficients ($\chi_l = g^l$), delta-M scaling, Nakajima-Tanaka corrections for intensities, ω_0 capped at $1-10^{-9}$ for conservative cloud particle scattering cases.
- All fluxes normalized to incident flux $\mu_0 \cdot F_0 = 1$.
- DISORT stream convergence: Reflectance stable to $\sim 10^{-5}$ from 16 to 64 streams; energy closure $R + T_{\text{net}} + A_{\text{cld}} = 1$ to 2×10^{-8}
- Other optical parameters: as specified in test cases

Verification performed on PythonicDISORT:

1. *Stream (angular) comparisons*: R for Case 1 varies by $< 2 \times 10^{-5}$ from NQuad = 16 to 64.
2. *Energy conservation*: $R + T_{\text{net}} = 1$ exactly (to 2×10^{-8}) for all $\omega_0 = 1$ cases; $F_{\uparrow} = A_{\text{sfc}} F_{\downarrow}$ at the base, confirming the implementation of the Lambertian boundary convention.
3. *Vs. independent Monte Carlo* (10^7 photons). Case 1: $R = 0.42202 \pm 0.00016$; Case 2: $R = 0.60408 \pm 0.00015$; Case 7: $R = 0.36258 \rightarrow \text{MC } 0.36241 \pm 0.00015$, $T_{\text{net}} = 0.25080$ vs 0.25087 . All within $\sim 1\sigma$.
4. *Intensity-flux closure*: hemispheric integration of the bidirectional intensities recovers the flux values to $\sim 0.5\%$ (expected level of trapezoid quadrature error at 24 polar nodes).

II. Flux and Absorption Comparisons

Fixed input parameters: Cloud Optical Thickness (COT) = 10, HG g = 0.85, Horizontal extent = 150 (optical path), Cloud-base to surface = 0.05 km. The latter two settings are to approximate a plane-parallel cloud layer for comparison with DISORT.

Output:

- Refl. Flux (R) = upward flux (albedo) at cloud-top
- Tran. flux down ($T\downarrow$) = total downward flux at cloud base
- Tran. flux net (T_{net}) = $F\downarrow(\tau) - F\uparrow(\tau)$ at base (also equal to flux absorbed by surface)
- Cloud absorptance (A_{cld}) = $1 - R - T_{\text{net}}$

DISORT v. Monte Carlo (100K photons unless noted)

Test Case	ω_0	μ_0	Surface albedo	R	$T\downarrow$	T_{net}	A_{cld}
1: DISORT	1.00	1.0	0.0	0.422	0.578	0.578	0.0
MC				0.422	0.578	0.578	0.0
MC (1M)				0.423	0.577	0.577	0.0
MC* (1M)				0.423	0.577	0.577	0.0
2: DISORT	1.00	0.5	0.0	0.604	0.396	0.396	0.0
MC				0.603	0.397	0.397	0.0
3: DISORT	0.98	1.0	0.0	0.287	0.402	0.402	0.311
MC				0.288	0.402	0.402	0.310
4: DISORT	0.98	0.5	0.0	0.449	0.251	0.251	0.300
MC				0.450	0.251	0.251	0.298
MC (1M)				0.449	0.250	0.250	0.301
MC* (1M)				0.449	0.250	0.250	0.300
5: DISORT	1.00	1.0	0.5	0.603	0.794	0.397	0.0
MC				0.600	0.797	0.400	0.0
6: DISORT	1.00	0.5	0.5	0.728	0.544	0.272	0.0
MC				0.727	0.544	0.273	0.0
7: DISORT	0.98	1.0	0.5	0.363	0.502	0.251	0.387
MC				0.364	0.503	0.252	0.385
8: DISORT	0.98	0.5	0.5	0.496	0.313	0.156	0.348
MC				0.496	0.316	0.157	0.347

* MC python code developed and run independent from the html/json code.

III. Bidirectional Direction Function (BDF) Comparisons

The BDF for a particular i,j bin direction ($BDF_{i,j}$) is given by $\pi I_{i,j}/(\mu_0 F_0)$, consistent with MC code's $(N_{i,j}/N_{\text{photons}}) \cdot \pi/(\mu_i \Delta\mu_i \Delta\phi_j)$. DISORT intensities are given at 24 up + 24 down quadrature angles $\times$ 72 relative azimuths, NT-corrected. Relative azimuth convention between the two codes is $\phi - \phi_0 = 0^\circ$ is the forward-scattering direction.

Case 1 ($\omega_0 = 1$, $\mu_0 = 1$, $A_{\text{sfc}} = 0$) with files *disort_case1_bdf_**

- DISORT field is azimuth-independent machine precision (only the $m = 0$ Fourier mode survives for $\mu_0 = 1$). Azimuthal speckle in the MC panels is consistent with Poisson noise as expected.
- Reflected BRF: nearly flat, 0.40–0.44, broad maximum near VZA $\approx 55^\circ$, falling to ~ 0.21 at the limb. Matches the MC's uniform green (~ 0.4 – 0.5) reflected panel.
- Transmitted BTF: ~ 0.82 at nadir decreasing monotonically to ~ 0.17 at the limb. Matches the MC's hotter center that fades to blue edges.

BDF polar plots

Photons: 100000, COT (τ): 10.00, Horizontal extent: 150.0

Θ_0 : 0.0° , HG g: 0.85, SSA (ω_0): 1.00

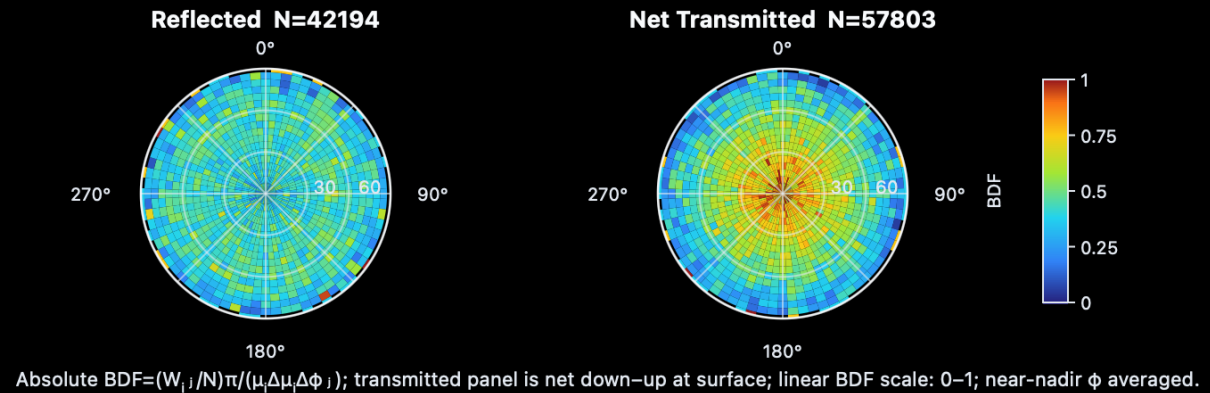
Surface Albedo A_s : 0.00, β_{ext} : 10.00 km^{-1}

d_{sfc} : 0.05 km, RNG seed: 42

R=0.422 (42194), T=0.578 (57803), A=0.000 (0), S=0.000 (3)

$A_{\text{sfc}}=0.000$ (0), $R+T+A+S+\text{Term}=1.000$

$T_{\text{down_sfc}}=0.578$ (57803), surface refl/photon=0.000 (0), Term (event cap)=0.000 (0)



MC with 100K photons, Case 1.

BDF polar plots

Photons: 1000000 , COT (τ): 10.00 , Horizontal extent: 150.0

Θ_0 : 0.0° , HG g: 0.85 , SSA (ω_0): 1.00

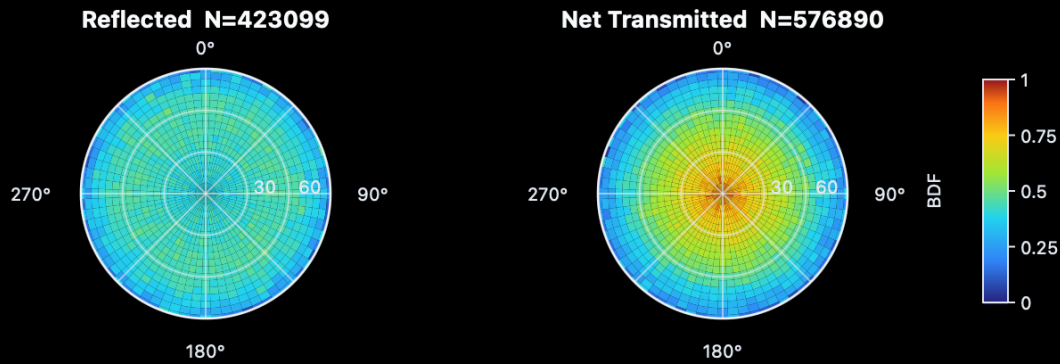
Surface Albedo A_s : 0.00 , β_{ext} : 10.00 km⁻¹

d_{sfc} : 0.05 km , RNG seed: 42

R=0.423 (423099) , T=0.577 (576890) , A=0.000 (0) , S=0.000 (11)

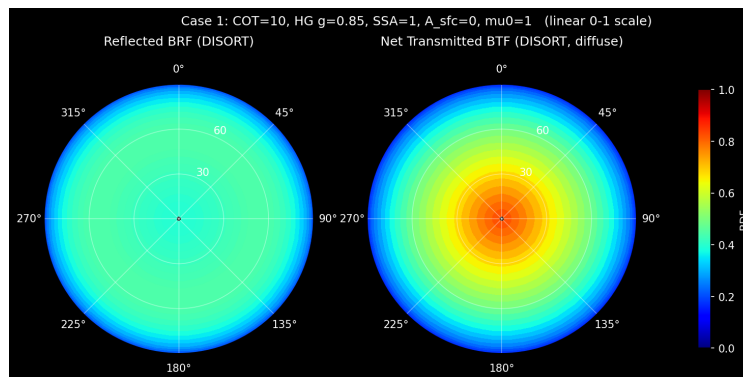
$A_{\text{sfc}}=0.000$ (0) , R+T+A+S+Term=1.000

$T_{\text{down_sfc}}=0.577$ (576890) , surface refl/photon=0.000 (0) , Term (event cap)=0.000 (0)



Absolute BDF= $(W_{ij}/N)\pi/(\mu_i\Delta\mu_j\Delta\phi_{ij})$; transmitted panel is net down-up at surface; linear BDF scale: 0–1; near-nadir ϕ averaged.

MC with 1M photons, Case 1.



DISORT, Case 1. [Note: not identical color bar v. MC]

Case 4 ($\omega_0 = 0.98$, $\mu_0 = 0.5$, $A_{\text{sfc}} = 0$) with files *disort_case4_bdf_**, *disort_case4_brf.csv*, *disort_case4_btf.csv*

- Integrated quantities from the 10^5 photon MC run (DISORT in parenthesis): $R = 0.449$ (0.44907), $T = 0.252$ (0.25053), $A = 0.299$ (0.30039), giving agreement to ~ 3 decimal points, which is consistent with expected MC noise.
- Reflected BRF: strong forward-scattering enhancement along $\phi = 0^\circ$, rising from ~ 0.3 at nadir to ~ 2.03 at $VZA \approx 86^\circ$, and exceeding 1.0 beyond $VZA \approx 60^\circ$ (clipped with the 0–1 color bar used in both figures). The angular extent of the MC's saturated dark-red region matches where DISORT crosses 1.0. Backward azimuth ($\phi = 180^\circ$) nearly flat at 0.26–0.31 since HG phase function as no glory/rainbow structure.
- Transmitted BTF: low values (0.06–0.35) and nearly azimuth-independent, slightly brighter at small VZA because multiple scattering for $\tau=10$ with significant absorption reduces solar-azimuth memory.
- Overall, MC results reproduce the BDF structure of the DISORT 48-stream solution, including the forward limb enhancement. Residual differences are visually consistent with Poisson noise ($\sim 25K$ transmitted photons spread over the angular bins).

BDF polar plots

Photons: 100000, COT (τ): 10.00, Horizontal extent: 150.0

Θ_0 : 60.0°, HG g: 0.85, SSA (ω_0): 0.98

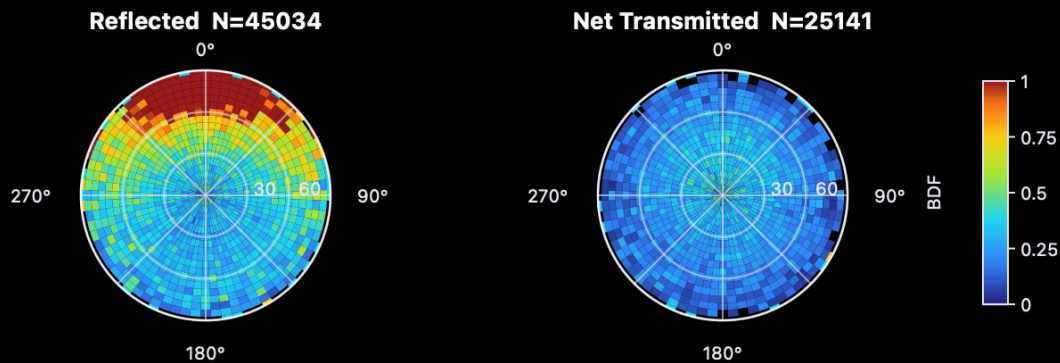
Surface Albedo A_s : 0.00, β_{ext} : 10.00 km $^{-1}$

d_{sfc} : 0.05 km, RNG seed: 42

R=0.450 (45034), T=0.251 (25141), A=0.298 (29825), S=0.000 (0)

$A_{\text{sfc}}=0.000$ (0), $R+T+A+S+\text{Term}=1.000$

$T_{\text{down_sfc}}=0.251$ (25141), surface refl/photon=0.000 (0), Term (event cap)=0.000 (0)



MC with 100K photons, Case 4.

BDF polar plots

Photons: 1000000 , COT (τ): 10.00 , Horizontal extent: 150.0

Θ_0 : 60.0° , HG g: 0.85 , SSA (ω_0): 0.98

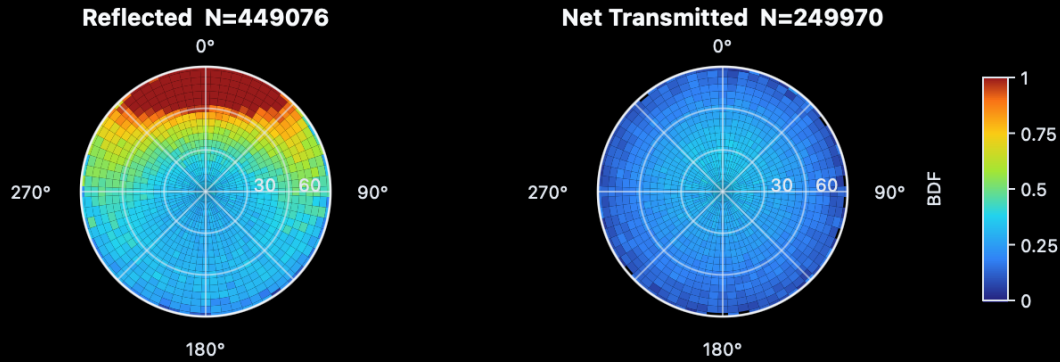
Surface Albedo A_s : 0.00 , β_{ext} : 10.00 km⁻¹

d_{sfc} : 0.05 km , RNG seed: 42

R=0.449 (449076) , T=0.250 (249970) , A=0.301 (300953) , S=0.000 (1)

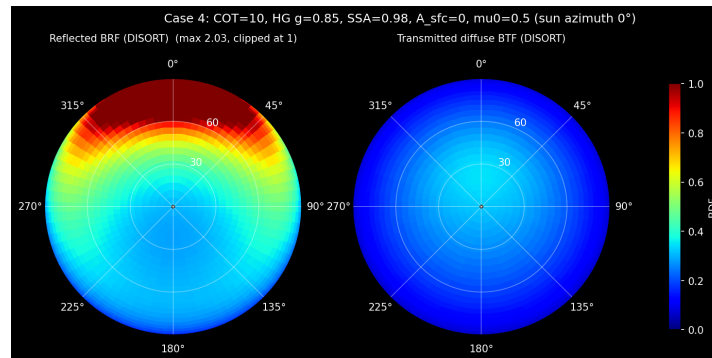
A_{sfc} =0.000 (0) , R+T+A+S+Term=1.000

$T_{\text{down_sfc}}$ =0.250 (249970) , surface refl/photon=0.000 (0) , Term (event cap)=0.000 (0)



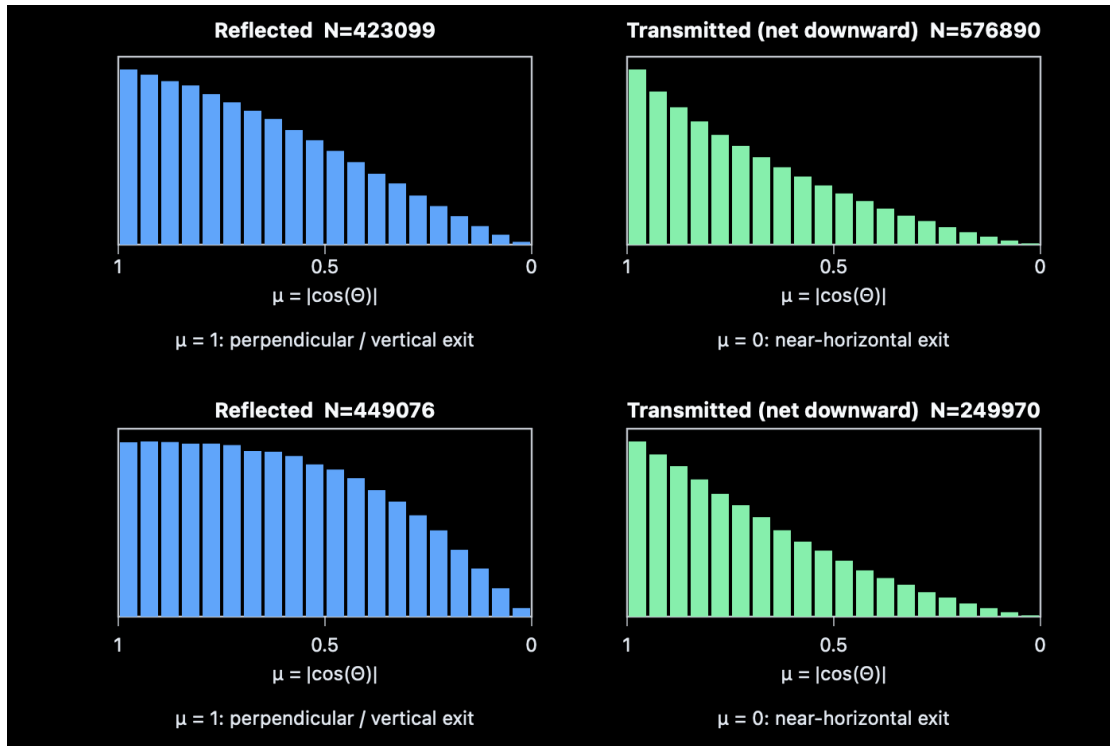
Absolute BDF= $(W_{ij}/N)\pi/(\mu_i\Delta\mu_j\Delta\phi_{ij})$; transmitted panel is net down-up at surface; linear BDF scale: 0–1; near-nadir ϕ averaged.

MC with 1M photons, Case 4.

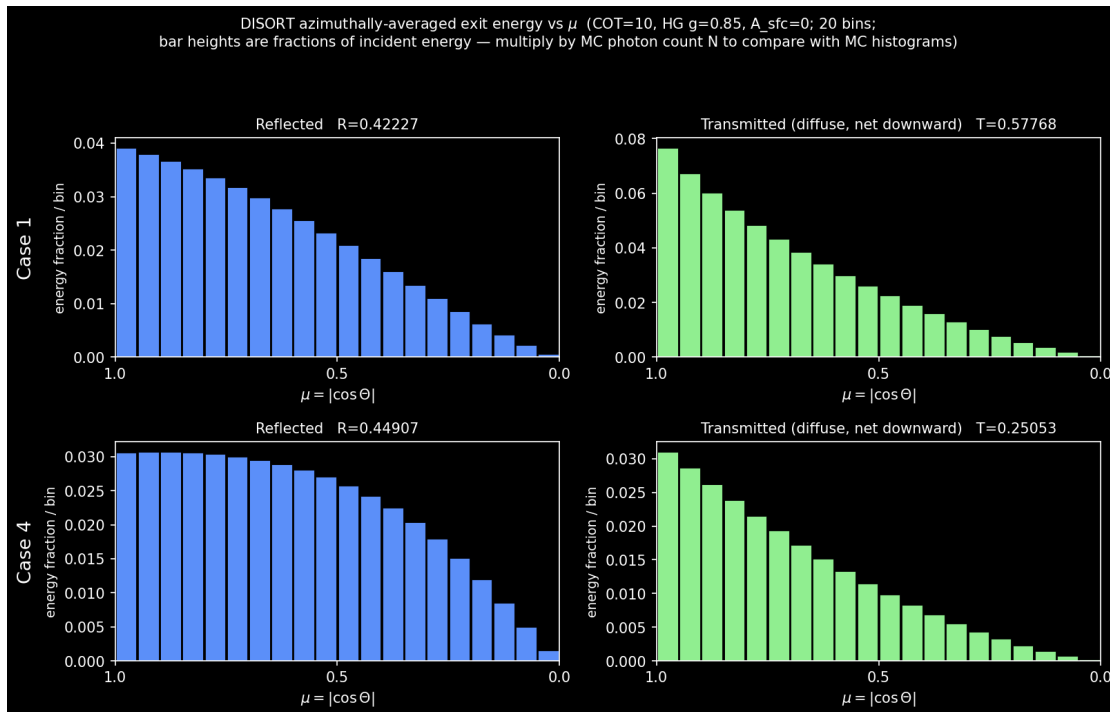


DISORT, Case 4. [Note: not identical color bar v. MC]

IV. View Angle (μ) Histograms

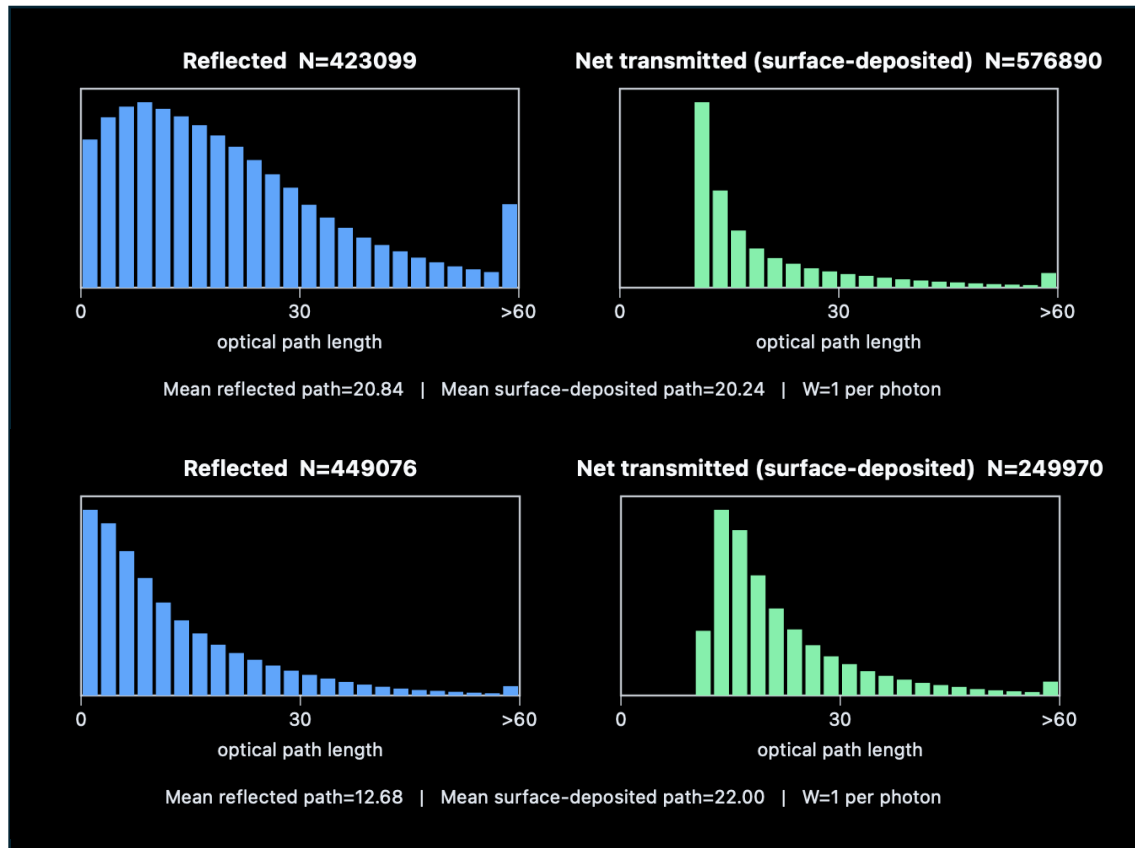


MC view angle histograms with 1M photons. Case 1 (top row) and Case 4 (bottom).



DISORT view angle histograms. Case 1 (top row) and Case 4 (bottom).

V. Other MC Output of Interest



MC path length histograms with 1M photons. Case 1 (top row) and Case 4 (bottom).

VI. Files in this test folder

- *disort_case_table.py*: 8 test cases script
- *disort_hg_test.py*: single-case flux script with stream-convergence loop
- *verify_hg_case.py*, *verify_table_mc.py*: delta-Eddington closed form and independent MC verification
- *disort_case1_bdf.py*, *disort_case4_bdf.py*: bidirectional intensity scripts
- *disort_case1_bdf_polar.png*, *disort_case1_bdf_vza.png*, *disort_case1_bdf.csv*
- *disort_case4_bdf_polar.png*, *disort_case4_bdf_pplane.png*, *disort_case4_brf.csv*, *disort_case4_btf.csv*